

**BEFORE YOU READ — ANSWER FROM MEMORY**

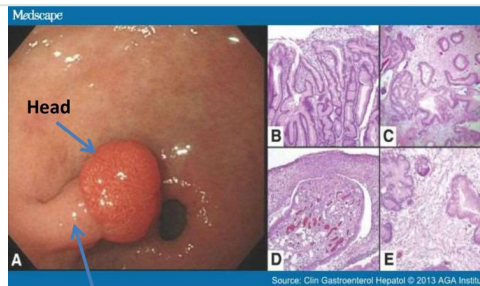
1. Which gastric polyp is **defined by dysplasia**, and what does that mean for its risk?
2. Which Lauren type has **no precursor lesion** and no known risk factors?
3. Name **three** metastatic eponyms of gastric carcinoma and their sites.

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Gastric polyps

A polyp is any nodule or mass projecting above the surrounding mucosa. Gastric polyps are uncommon (compared with colonic).

Type	Association	Malignant potential
Hyperplastic / inflammatory	Commonest ; arise in chronic gastritis	—
Fundic gland	FAP and chronic PPI use	—
Adenomatous	Defined by dysplasia	Definite risk of harbouring adenocarcinoma



Hyperplastic polyp on a stalk in the antrum (head, stalk)

2 Gastric cancer — the frame

Commonest	Adenocarcinoma >90% of gastric cancers.
Lymphoma	The stomach is the commonest site of extranodal NHL .
Location	Most involve the antrum ; lesser curvature more than greater curvature.
Geography	Incidence varies markedly. Rates fell >85% in the 20th century: less dietary N-nitroso compounds and benzo[a]pyrene (less salting and smoking of food), plus mass endoscopic screening in high-incidence regions detecting early cancers.
Presentation	Abdominal discomfort · early satiety · nausea and vomiting · GI hemorrhage, iron-deficiency anemia.

THE BLOCK'S RECURRING PATTERN

Chronic injury → metaplasia → dysplasia → carcinoma. In the stomach: **H. pylori** → **chronic gastritis** → **gastric atrophy** → **intestinal metaplasia** → **epithelial dysplasia** → **carcinoma**.

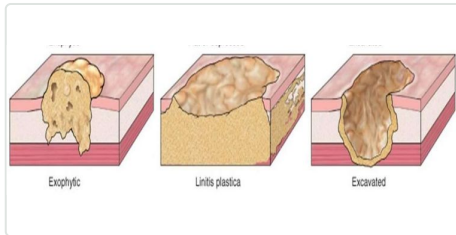
Risk factors (multifactorial)

Infection	H. pylori, via the sequence above
Predisposing conditions	Atrophic gastritis, autoimmune gastritis, pernicious anemia · adenomatous polyps · subtotal gastrectomy · gastric ulcer
Lifestyle / social	Smoking · lower socioeconomic groups · nitrosamine
Diet: increases risk	Large amounts of starch; smoked or cured fish and meat; pickled vegetables
Diet: decreases risk	Whole milk; fresh vegetables rich in vitamin C and other antioxidants

3 Morphology — gross and Lauren classification

Gross patterns	Polypoidal (exophytic) · infiltrative (linitis plastica) · ulcerative (excavated) .
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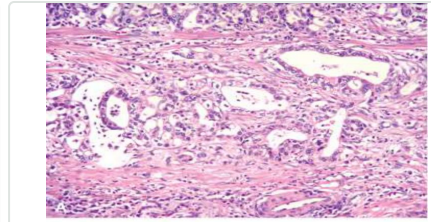
	Intestinal type	Diffuse type
Histology	Glandular structures , like esophageal and colonic adenocarcinoma	Signet ring cell morphology
Growth	Exophytic mass or ulcerated tumour	Infiltrative; desmoplastic reaction stiffens the wall, flattens rugae: "leather bottle" = linitis plastica
Precursor / risk	Usually arises from a precursor lesion ; has risk factors	No precursor lesion, no known risk factors



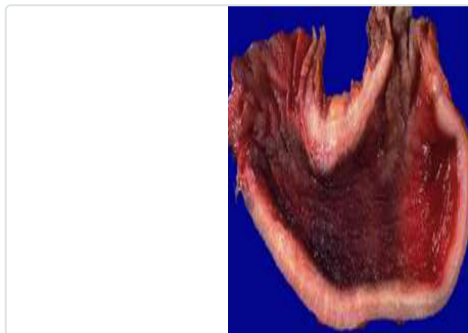
Exophytic · linitis plastica · excavated



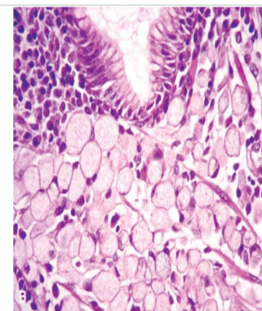
Intestinal type: excavated ulcer, heaped-up borders, dirty base



Intestinal type: glands invading muscle



Diffuse type: linitis plastica

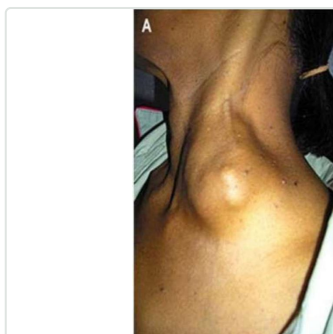


Diffuse type: signet ring cells

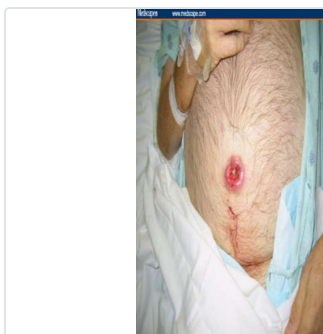
4 Metastases — the eponyms

Often detected at the time of diagnosis.

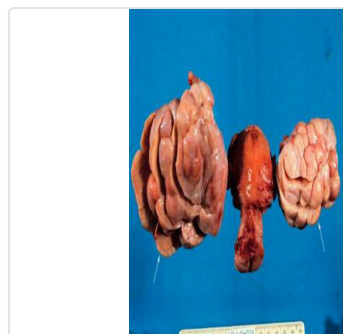
Site	Eponym
Supraclavicular lymph node	Virchow node / Troisier sign
Periumbilical region	Sister Mary Joseph nodule
Left axillary lymph node	Irish node
Ovary	Krukenberg tumour
Pouch of Douglas	Blumer shelf



Troisier sign (supraclavicular node)



Sister Mary Joseph nodule



Krukenberg tumour

P Must-remember pearls

- **Adenocarcinoma is >90% of gastric cancers**; the stomach is also the commonest site of extranodal NHL.
- **H. pylori → gastritis → atrophy → intestinal metaplasia → dysplasia → carcinoma**: learn the chain, not the endpoints.
- **Intestinal type has a precursor and risk factors; diffuse type has neither** and shows signet ring cells with linitis plastica.
- **Adenomatous polyps are defined by dysplasia** — the reason they carry definite cancer risk.
- **Krukenberg = ovary; Virchow = supraclavicular; Sister Mary Joseph = umbilicus; Irish = left axilla; Blumer = pouch of Douglas.**

Q Test yourself

1. A gastrectomy specimen shows a diffusely thickened, rigid stomach wall with flattened rugae. Histology shows signet ring cells. Which statement is correct?

- A.** It has no known precursor lesion or risk factors
- B.** It typically arises from a recognised precursor lesion
- C.** It forms glands resembling colonic adenocarcinoma
- D.** It usually presents as an exophytic mass
- E.** It is the intestinal type

2. A woman with gastric adenocarcinoma is found to have ovarian masses. What are they called?

- A.** Blumer shelf
- B.** Irish node
- C.** Sister Mary Joseph nodule
- D.** Krukenberg tumour
- E.** Virchow node

3. Which gastric polyp carries a definite risk of harbouring adenocarcinoma?

- A.** Hyperplastic polyp
- B.** Fundic gland polyp
- C.** Adenomatous polyp
- D.** Inflammatory polyp
- E.** All gastric polyps equally

4. Gastric adenocarcinoma incidence fell by more than 85% in the twentieth century. Which factor is listed as a possible contributor?

- A.** Increased use of PPIs
- B.** Wider use of subtotal gastrectomy
- C.** Rising consumption of pickled vegetables
- D.** Increased starch intake
- E.** Reduced dietary N-nitroso compounds and benzo[a]pyrene as salting and smoking of food declined

5. Endoscopy shows a gastric ulcer with irregular, heaped-up borders and a dirty base. Compared with a benign peptic ulcer, which feature is most suggestive of malignancy?

- A.** Regular margins
- B.** Heaped-up irregular borders
- C.** Radiating mucosal folds
- D.** A clean, smooth base
- E.** A solitary lesion

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 • A — Linitis plastica with signet ring cells is the diffuse type: no known precursor lesion or risk factors, the opposite of the intestinal type.

2 • D — Ovarian metastasis of gastric carcinoma is a Krukenberg tumour.

3 • C — Adenomatous polyps are defined by dysplasia and have a definite risk of harbouring adenocarcinoma.

4 • E — Less use of salt and smoking for food preservation cut intake of N-nitroso compounds and benzo[a]pyrene; mass endoscopic screening also detected early cancers.

5 • B — Benign peptic ulcers have regular margins, radiating folds and a clean base; the intestinal-type carcinoma ulcer has heaped-up borders and a dirty base.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Study alongside the lecture, not instead of it.